

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 5: System Software

Subtopic 5.2: Language Translators

Concept 5.2.2: Comparison of Compilers and Interpreters

Total Questions: 13

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_12 (Q10.a)

10 A programmer is developing a computer program.

- (a) Explain how the programmer can first use an interpreter **and** then a compiler to develop the computer program.

Interpreter

.....

.....

.....

.....

Compiler

.....

.....

.....

[4]

Question 3 | Source: 9618_s25_qp_13 (Q3.a)

3 A programmer is writing a program in a high-level language.

- (a) Complete the description of compilers and interpreters by writing the missing words.

A compiler checks all the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces which can run without access to the

An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter and displays the error. The programmer can correct the error and then the interpreter continues translating from that point.

[4]

Question 4 | Source: 9618_s24_qp_11 (Q3.a)

3 A software developer is writing a computer program.

- (a) The developer uses an interpreter while writing the program code because it is easier for debugging.

Explain **one** reason why it is easier to debug the program code using an interpreter instead of a compiler.

.....

.....

.....

..... [2]

Question 5 | Source: 9618_s24_qp_11 (Q3.b)

- 3 A software developer is writing a computer program.

- (b) The program is ready to be sold to customers.

The developer uses a compiler because it creates an executable file.

Explain the reasons why the need to create an executable file makes the compiler the appropriate choice when the program is complete.

.....

.....

.....

.....

.....

..... [3]

Question 6 | Source: 9618_s24_qp_12 (Q8.a)

- 8 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

- (a) The programmer decides to use the compiler when testing the final program.

Describe the benefits of using the compiler during testing.

.....

.....

.....

..... [2]

Question 7 | Source: 9618_w23_qp_11 (Q6.a)

6 A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

(a) Describe the advantages of using the interpreter compared to the compiler to translate the program.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 8 | Source: 9618_w23_qp_11 (Q6.b)

6 A programmer uses both a compiler and an interpreter to translate a program written in a high-level language.

(b) State **one** reason why some high-level languages are partially compiled and partially interpreted.

.....

..... [1]

Question 9 | Source: 9618_s23_qp_12 (Q7.b)

7 A software developer is working in a team writing a program for a client.

- (b) The development team needs to use a translator whilst writing the program for the client.

Identify whether an interpreter or a compiler would be more appropriate at this stage of the program development.

Justify your choice.

Translator

Justification

.....

.....

.....

.....

..... [3]

Question 10 | Source: 9618_s23_qp_13 (Q5.a.i)

- 5 A programmer is developing a computer game in a high-level language to sell to the public.

- (a) The programmer uses both an interpreter and a compiler at different stages of the development of the program.

- (i) Explain the reasons why the programmer uses an interpreter while writing the program code.

.....

.....

.....

..... [2]

Question 11 | Source: 9618_s23_qp_13 (Q5.a.ii)

- 5 A programmer is developing a computer game in a high-level language to sell to the public.

- (a) The programmer uses both an interpreter and a compiler at different stages of the development of the program.

- (ii) Explain the reasons why the programmer uses a compiler when the program has been written.

.....

.....

.....

.....

.....

..... [3]

Question 12 | Source: 9618_w22_qp_12 (Q1.b.i)

- (b) Compilers and interpreters translate programs written in a high-level language into a low-level language.

- (i) State **two** drawbacks of using a compiler compared to an interpreter during program development.

1

.....

.....

2

.....

.....

[2]

Question 13 | Source: 9618_w22_qp_12 (Q1.b.ii)

- (b) Compilers and interpreters translate programs written in a high-level language into a low-level language.

- (ii) Explain why high-level language programs might be partially compiled and partially interpreted.

.....

.....

.....

..... [2]

Question 14 | Source: 9618_s21_qp_12 (Q7.b.i)

- 7 Jennifer is writing a computer program for her A Level homework.

- (b) Jennifer uses an Integrated Development Environment (IDE) to write her computer program.

- (i) The IDE allows Jennifer to use both an interpreter and a compiler while creating her computer program.

Describe the ways in which Jennifer can use **both** a compiler **and** an interpreter while developing the program.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q10.a)

10(a)	1 mark per bullet point, max 4 marks Max 2 from each section. Interpreter: • Use an interpreter while writing/coding the program • ... to test / debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file so source code not seen • Use the compiler to repeatedly test the same completed section without having to re-interpret every time	4
-------	---	----------

Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q8.a)

8(a)	1 mark per bullet point, max 2 • Large amounts of source code take time to compile • It can be slower to produce the object code than an interpreter • The code must be recompiled when it is changed • The program cannot run if there are errors • It is not possible to correct errors in real-time • One error can cause false reporting of multiple further errors • Sections of code / unfinished code cannot easily be tested	2
------	---	----------

Mark Scheme for Question 3 | Source: 9618_s25_ms_13 (Q3.a)

3(a)	1 mark for each correctly completed term • an executable file/.exe • source/program code • stops • immediately/in real-time A compiler checks all of the code before attempting to translate the program. If any errors are found, they are all reported at the same time and the program does not translate or run. If there are no errors found, the compiler produces an executable file/.exe which can run without access to the source/program code. An interpreter translates one line of code and then runs it, before moving to the next line of code. If the line of code has an error, the interpreter stops and displays the error. The programmer can correct the error immediately/in real-time and then the interpreter continues translating from that point.	4
------	---	----------

Mark Scheme for Question 4 | Source: 9618_s24_ms_11 (Q3.a)

3(a)	1 mark each to max 2: • The interpreter will stop when an error is found • ... so the error can be corrected in real-time, and the result of changes seen immediately • Only one error is displayed at a time • ... so fewer errors to correct simultaneously and no dependent errors	2
------	--	----------

Mark Scheme for Question 5 | Source: 9618_s24_ms_11 (Q3.b)

3(b)	1 mark each to max 3: • Program can be distributed without source code • ... so it cannot be edited/stolen/plagiarised • Users do not require the translator to run the program • ... so time is not spent retranslating by user	3
------	--	----------

Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q8.a)

8(a)	1 mark each to max 2: • Creates an executable file • ... so the code can be tested multiple times without having to recompile • ... so repeated testing takes less time	2
------	---	----------

Mark Scheme for Question 7 | Source: 9618_w23_ms_11 (Q6.a)

6(a)	1 mark for each bullet point (max 4) - easier to debug the program ... because it translates line-by-line and stops when an error is found whereas the compiler translates all the program at the same time ... only reporting one error at a time ... which allows the error to be corrected in real time whereas the program would need to be corrected and recompiled ... and the program can restart at same point when error occurred with a compiler the program needs to be re-run - The effect of any changes made by the programmer can be seen immediately with a compiler the effects can only be seen after re-running - A partially completed program can be translated / tested on its own a compiler cannot translate a partial program	4
------	---	----------

Mark Scheme for Question 8 | Source: 9618_w23_ms_11 (Q6.b)

6(b)	1 mark for each bullet point (max 1) - Partially compiled programs can be used on different platforms as they are interpreted when run - Code is optimised for the CPU as machine code is generated at run time - Source code does not need recompiling so more efficient to run	1
------	---	----------

Mark Scheme for Question 9 | Source: 9618_s23_ms_12 (Q7.b)

7(b)	No mark for choice. 1 mark each to max 3 for justification Interpreter - Allows the developer to make real-time changes ... so the program can be debugged at each stage ... the effect of any changes made by the developer can be seen immediately - The developer can test when incomplete ... so small parts can be tested without having to test the rest of the program ... if one section does not work others can still be tested - To avoid dependent errors Compiler - The developer can debug multiple errors simultaneously - Produces an executable file ... so that the developer can test the program multiple times without recompiling	3
------	---	----------

Mark Scheme for Question 10 | Source: 9618_s23_ms_13 (Q5.a.i)

5(a)(i)	1 mark each to max 2 • Programmer can test sections of the code without every part working / being written • Programmer can debug in real time • ... so that errors can be fixed and the program continued from that point • The effect of any changes made by the programmer can be seen immediately • To avoid dependent errors	2
---------	---	----------

Mark Scheme for Question 11 | Source: 9618_s23_ms_13 (Q5.a.ii)

5(a)(ii)	1 mark each to max 3 • The compiler produces an executable file • ... so the user cannot access / edit / sell the code • ... and users do not need the translator to run the game • The game can be compiled for different hardware specifications • ... and then used to generate more income for the programmer • The program can be tested multiple times without having to retranslate each time	3
----------	---	----------

Mark Scheme for Question 12 | Source: 9618_w22_ms_12 (Q1.b.i)

1(b)(i)	1 mark for each bullet point (max 2): • larger amounts of source code take time to compile • slower to produce the object code than an interpreter • code cannot be changed without recompilation • the program will not run if there are any errors • errors cannot be corrected in real-time • one error may result in other false errors being reported • cannot easily test specific sections of the source code // cannot easily test unfinished source code	2
---------	---	----------

Mark Scheme for Question 13 | Source: 9618_w22_ms_12 (Q1.b.ii)

1(b)(ii)	1 mark for each bullet point (max 2): • partially compiled programs can be used on different platforms as they are interpreted when run • code is optimised for the CPU as machine code is generated at run time	2
----------	--	----------

Mark Scheme for Question 14 | Source: 9618_s21_ms_12 (Q7.b.i)

7(b)(i)	1 mark per bullet point to max 4; max 3 from each section Interpreter: • Use an interpreter while writing the program • ... to test/debug the partially completed program • ... because errors can be corrected and processing continue from where the execution stopped // errors can be corrected in real time // errors are identified one at a time Compiler: • Use the compiler after the program is complete • ... to create an executable file • Use the compiler to repeatedly test the same (completed) section • ... without having to re-interpret every time // compiler not needed at run-time	4
---------	---	----------